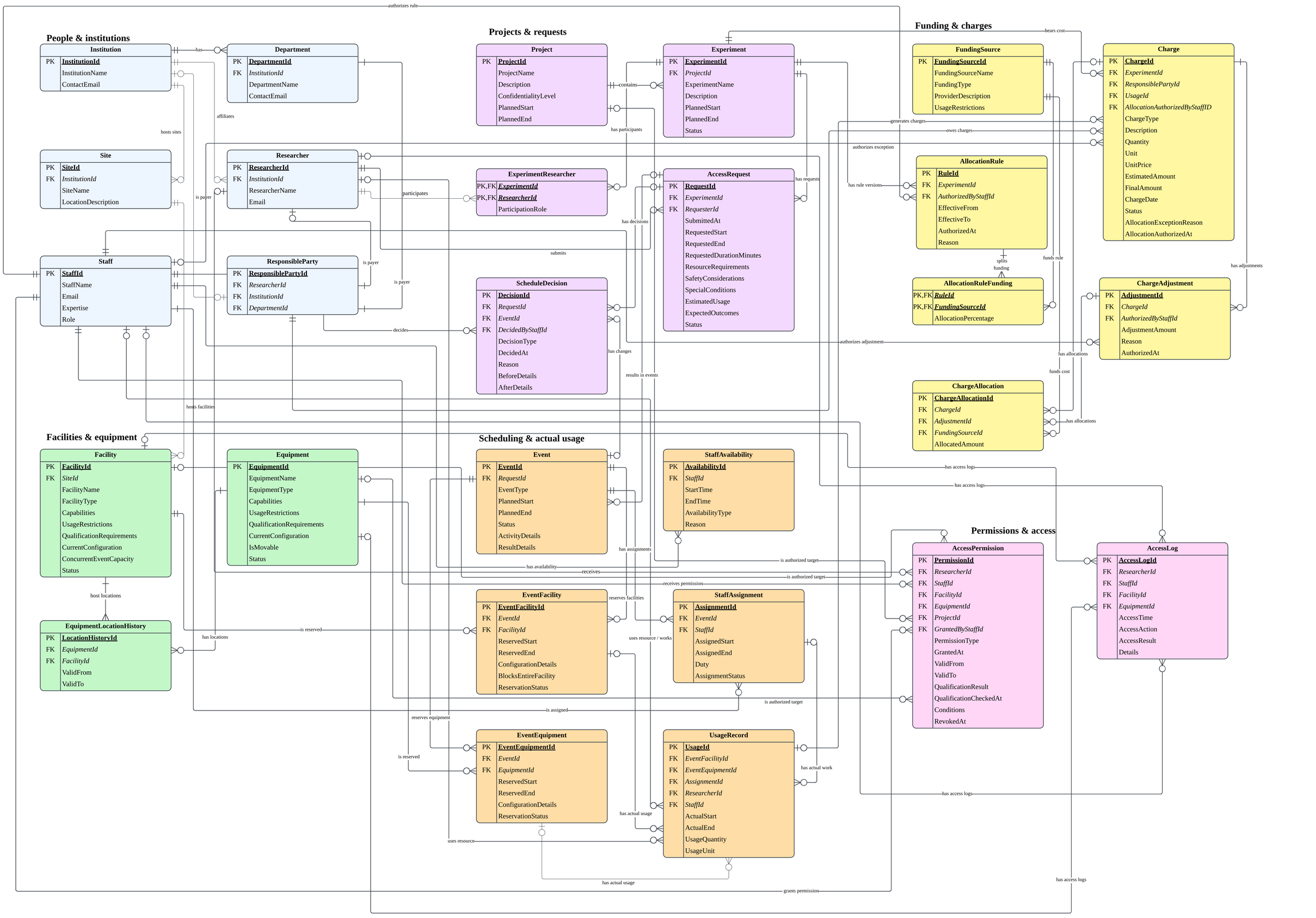


Assumptions:

1. payer identity

ResponsibleParty: exactly one of ResearcherId, InstitutionId or DepartmentId is not mull. Each not null identity is unique (one payer record per person / institution / department).

2. decision subject

ScheduleDecision: exactly one of RequestId or EventId is not nulk. Request decisions preserve the submitted application; Event decisions preserve scheduling, resource, configuration and staff changes.

3. actual usage

Exactly one source: EventFacilityId / EventEquipmentId / AssignmentId. For staff-work rows (AssignmentId present), ResearcherId and StaffId are null; actual staff is obtained from StaffAssignment.StaffId. For resource-use rows, exactly one actor: ResearcherId / StaffId.

4. permission scope

AccessPermission: exactly one actor (ResearcherId / StaffId) and exactly one target (FacilityId / EquipmentId / ProjectId). Resource-use permission targets a facility or equipment; information-view permission targets a project. GrantedByStaffId is required.

5. logged access

AccessLog: exactly one actor (ResearcherId / StaffId) and exactly one resource (FacilityId / EquipmentId). Logs record resource access, including denied attempts; a failed attempt need not have a valid permission.

6. cost allocation

ChargeAllocation: exactly one of ChargeId / AdjustmentId is non-NULL. FundingSourceId is required. Each source appears at most once per charge or adjustment. Final allocation totals equal FinalAmount or AdjustmentAmount.